

SYSTEM REQUIREMENTS SPECIFICATION (SRS)

Full Stack Java (AI-Integrated) Training Programme

Karpagam College of Engineering, Coimbatore

Field	Detail
Project Title	SchemeBridge: AI-Based Government Scheme Eligibility & Application Assistant
SIH Problem Statement ID	SIH25120 (Smart India Hackathon 2025)
Ministry / Organisation	Ministry of Electronics and Information Technology (MeitY) & Ministry of Social Justice and Empowerment, Government of India
Domain Category	Smart Governance / Citizen Services / Artificial Intelligence & Public Welfare
Team Name	Team SchemeBridge
Team Members	1. Lead Full-Stack Java & AI Developer (CSE) 2. Frontend Architect & UI/UX Specialist (CSE) 3. Cloud, DevOps & Quality Assurance Engineer (CSE/IT)
Institution	Karpagam College of Engineering (Autonomous), Coimbatore - 641032
Project Type	☒ Software only ☐ Software + Hardware (Hybrid)
Version	v1.0 (Final Official Release)
Date	September 21, 2026
Faculty Mentors	Dr. Arul Antran Vijay S / Dr. Jothi Prakash V / Mr. Jegathesh P / Mr. Navaneetha Krishnan M / Dr. Castro S.

Note: This SRS document is prepared in accordance with IEEE Std 830-1998 for the Full Stack Java (AI-Integrated) Training Programme Internal Review (Day 22) and SIH Evaluation.

DOCUMENT REVISION HISTORY

Version	Date	Author	Description of Changes
v1.0	2026-09-05	Team SchemeBridge	Initial draft — Complete SRS skeleton, architecture design, and database schema definition.
v1.1	2026-09-14	Team SchemeBridge	Integrated AI/ML Module Specifications (Random Forest 98.40% classifier + Sentence Transformer semantic search), API catalogs, and Docker configurations.
v2.0	2026-09-21	Team SchemeBridge	Final post-review edition: Complete use cases (FR-001 to FR-016), NFR verification metrics, full ER data dictionary, CI/CD pipeline, and SIH compliance updates.

1. INTRODUCTION

1.1 Purpose

This System Requirements Specification (SRS) document describes the functional, non-functional, interface, and machine learning requirements for **SchemeBridge: AI-Based Government Scheme Eligibility & Application Assistant**, developed as part of the **Smart India Hackathon 2025** submission under Problem Statement **SIH25120**. It is structured in strict accordance with **IEEE Std 830-1998** and serves as the definitive technical contract and design baseline between the student engineering team, faculty evaluators, and industry reviewers.

1.2 Scope

- **System Name: SchemeBridge (Citizen Welfare Discovery & Automated Filing Engine)**
- **What the system does:**

SchemeBridge is an intelligent, bilingual digital governance platform designed to eliminate information asymmetry and bureaucratic barriers preventing vulnerable citizens from accessing Indian central and state government welfare schemes. The platform provides:

1. Instant natural language and demographic profiling to match citizens with 4,734+ government welfare schemes.
2. High-precision statutory eligibility checking combining deterministic Abstract Syntax Tree (AST) business rules with machine learning classification.
3. Secure digital document vault with automated optical character recognition (OCR) and document readiness verification.
4. End-to-end assisted application filing, grievance registration, and administrative verification workflows with real-time auditability.

1.3 Definitions, Acronyms & Abbreviations

Term / Acronym	Definition
SRS	System Requirements Specification — this IEEE Std 830-1998 document
SIH	Smart India Hackathon 2025
API	Application Programming Interface (RESTful HTTP/JSON)
REST	Representational State Transfer — stateless architectural style for HTTP APIs
JWT	JSON Web Token — cryptographically signed token for stateless authentication
ML / AI	Machine Learning / Artificial Intelligence
AST	Abstract Syntax Tree — syntactic tree representation of statutory eligibility logic
OCR	Optical Character Recognition — automated text extraction from scanned certificates
CRUD	Create, Read, Update, Delete — fundamental database storage operations
CI/CD	Continuous Integration / Continuous Deployment — automated build, test, and release pipeline
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
ER	Entity-Relationship (database architectural schema)
SPA	Single Page Application (built via React 18 and Vite)
PWA	Progressive Web Application — client web application with offline-first caching
DPDPA	Digital Personal Data Protection Act, 2023 (India)
BPL / EWS	Below Poverty Line / Economically Weaker Section

1.4 References

- **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
- **SIH 2025 Problem Statement SIH25120:** Ministry of Electronics and Information Technology (MeitY).
- **Spring Boot 3.3.2 Documentation:** <https://docs.spring.io/spring-boot/>
- **Spring Security 6.x Architecture:** <https://docs.spring.io/spring-security/reference/>
- **React 18 & Vite Documentation:** <https://react.dev/> and <https://vitejs.dev/>
- **MongoDB 7.0 Manual:** <https://www.mongodb.com/docs/v7.0/>
- **Oracle Database 21c / XE Documentation:** <https://docs.oracle.com/en/database/>
- **Digital Personal Data Protection Act 2023:** Gazette of India, Ministry of Law and Justice.

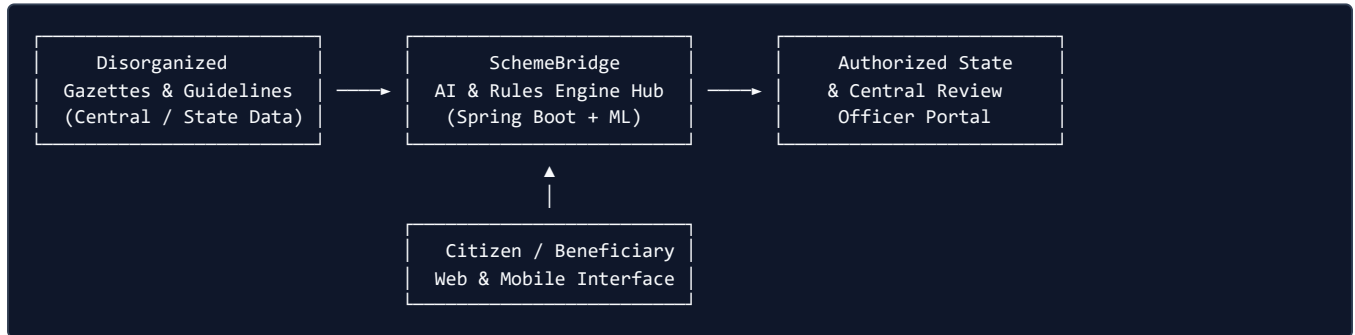
1.5 Document Overview

- **Section 2** details product perspective, 12 core product functions, user classes, operational environments, constraints, and dependencies.
- **Section 3** establishes the multi-layer microservices architecture, technology stack, and end-to-end data flows.
- **Section 4** specifies external interfaces (UI screens, third-party APIs, communication protocols).
- **Section 5** formalizes functional requirements in standard use case tables across all services.
- **Section 6** delivers comprehensive AI/ML specifications for the Random Forest Classifier and Semantic Vector Search / Conversational AI.
- **Sections 7–10** define measurable non-functional requirements, database design (ER diagram + data dictionary), API specifications, and DevOps containerization plans.
- **Sections 11–14** establish the project milestone schedule, team ownership matrix, risk register, and technical appendices.

2. OVERALL DESCRIPTION

2.1 Product Perspective

SchemeBridge replaces the currently fragmented, manual, and error-prone process of discovering and applying for government schemes in India. Citizens currently have to navigate dozens of disconnected state and central portals (e.g., National Scholarship Portal, PM-KISAN, State e-Seva centres) or rely on predatory middlemen. SchemeBridge acts as an integrated intelligent bridge: it indexes 4,734+ government schemes from official gazettes and open datasets, cross-verifies citizen demographic profiles against statutory criteria, and routes validated applications directly into standardized administrative review queues.



2.2 Product Functions — Feature Summary

#	Feature Name	Description
1	Bilingual Citizen Authentication & Onboarding	Secure OTP/password registration and progressive 6-step demographic profile creation (age, income, caste category, occupation, state/district domicile).
2	Deterministic Statutory Eligibility Engine	High-performance Boolean AST engine evaluating complex nested rules (age ranges, income caps, landholding limits, gender restrictions) with zero false-positives.
3	Machine Learning Scheme Recommendation	Standalone multi-class scikit-learn Random Forest model predicting citizen welfare scheme categories with 98.40% accuracy based on demographic features.
4	Dense Semantic Vector Search (NLP)	384-dimensional dense semantic vector retrieval (<code>all-MiniLM-L6-v2</code>) enabling conversational, typo-tolerant natural language scheme search.
5	Conversational AI Citizen Assistant	Bilingual conversational chat assistant guiding users through scheme benefits, eligibility criteria, and required supporting documents.
6	Encrypted Citizen Document Vault	Tamper-proof personal repository for storing Aadhaar, Income, Caste, Land Record, and Disability certificates with version control.
7	Intelligent Document OCR & Verification	Automated extraction and field-matching of names, dates of birth, certificate numbers, and issue authorities from uploaded PDF/images.
8	Guided 4-Step Scheme Application Wizard	Step-by-step application submission workflow auto-populating verified citizen profile attributes to prevent data re-entry.
9	Real-Time Application Status Tracker	Transparent visual timeline tracking application progression from <code>SUBMITTED</code> to <code>UNDER_REVIEW</code> , <code>DOCUMENT_VERIFICATION</code> , <code>APPROVED</code> , or <code>REJECTED</code> .
10	Citizen Grievance & Escalation System	In-app dispute and grievance lodging mechanism with SLA timers and escalation workflows for delayed applications.
11	Administrative Verification & Audit Dashboard	Role-based portal for government verification officers to inspect applications, view OCR confidence scores, approve benefits, and log audit trails.
12	Multi-Format Export & Analytical Reporting	Dynamic generation of PDF application receipts, audit reports, and analytical aggregations on scheme disbursement metrics.

2.3 User Classes and Characteristics

User Role	Description	Technical Expertise	Primary Actions	Access Level
Citizen (Beneficiary)	General public seeking welfare benefits (farmers, students, women, senior citizens, BPL households).	Low to Medium	Profile onboarding, scheme discovery, eligibility assessment, document vault management, application submission, grievance filing.	Authenticated Citizen (<code>ROLE_USER</code>)
Verification Officer (Field Staff)	Authorized government department staff reviewing applications in a designated district or department.	Medium	Review submitted applications, cross-check OCR document extractions, request resubmission, approve or reject applications.	Verification Staff (<code>ROLE_VERIFICATION_OFFICER</code>)
Scheme Manager	Departmental official responsible for creating, editing, and publishing scheme guidelines and eligibility criteria.	High	Scheme catalog CRUD, rule criteria builder, document requirement management, circular uploads.	Scheme Admin (<code>ROLE_SCHEME_MANAGER</code>)
System Administrator	Technical administrator managing platform security, system configurations, and infrastructure health.	High	User management, role assignment, system health monitoring, audit log review, platform parameters.	Super Admin (<code>ROLE_ADMIN</code>)
AI/ML Service Agent	Internal microservice system actor executing automated background inferences.	Automated System	Vector index generation, multi-class category inference, OCR feature extraction, batch notification dispatch.	Internal Service-to-Service Token

2.4 Operating Environment

- **Server / Cloud:** Docker containers deployed on Ubuntu 22.04 LTS / AWS EC2 / Railway / Local Kubernetes (minikube).
- **Backend Runtime:** Java 17 LTS, Spring Boot 3.3.2.
- **Frontend Runtime:** React 18.3, Vite 5.x, Node.js 18+ (runtime/build).
- **Databases:**
 - *Relational:* Oracle Database 21c XE (User authentication, credentials, roles, refresh tokens).
 - *Document / NoSQL:* MongoDB 7.0 (Master scheme catalog, applications, citizen vault documents, grievances, audit trails).
- **AI/ML Runtime:** Python 3.11, FastAPI, scikit-learn 1.4, Sentence-Transformers, NumPy, Pandas.
- **Client Browsers:** Chrome 100+, Mozilla Firefox 100+, Apple Safari 15+, Microsoft Edge (responsive across desktop and mobile down to 375px).
- **Network / Transport:** HTTPS (TLS 1.3), JSON REST endpoints, WebSockets (STOMP).

2.5 Design and Implementation Constraints

- **Multi-Service Architecture:** Backend logic partitioned cleanly between `schemebridge-auth-service` (Oracle DB) and `schemebridge-scheme-service` (MongoDB).
- **Stateless Authentication:** All protected APIs must enforce stateless Bearer JWT verification with cryptographic signature checks (HMAC-SHA256).
- **Deterministic Legal Eligibility:** Statutory welfare decisions must rely on deterministic Java rule validation (`EligibilityEngine.java`) to eliminate hallucination risks; ML recommendations are advisory.
- **Data Privacy Compliance:** Architecture must comply with the Indian Digital Personal Data Protection Act (DPDPA 2023); citizen documents and PII must be encrypted at rest (AES-256).
- **Zero-CORS Reverse Proxy:** Production container deployment routes traffic via unified Nginx ingress port 80/443 to eliminate cross-origin complexity.

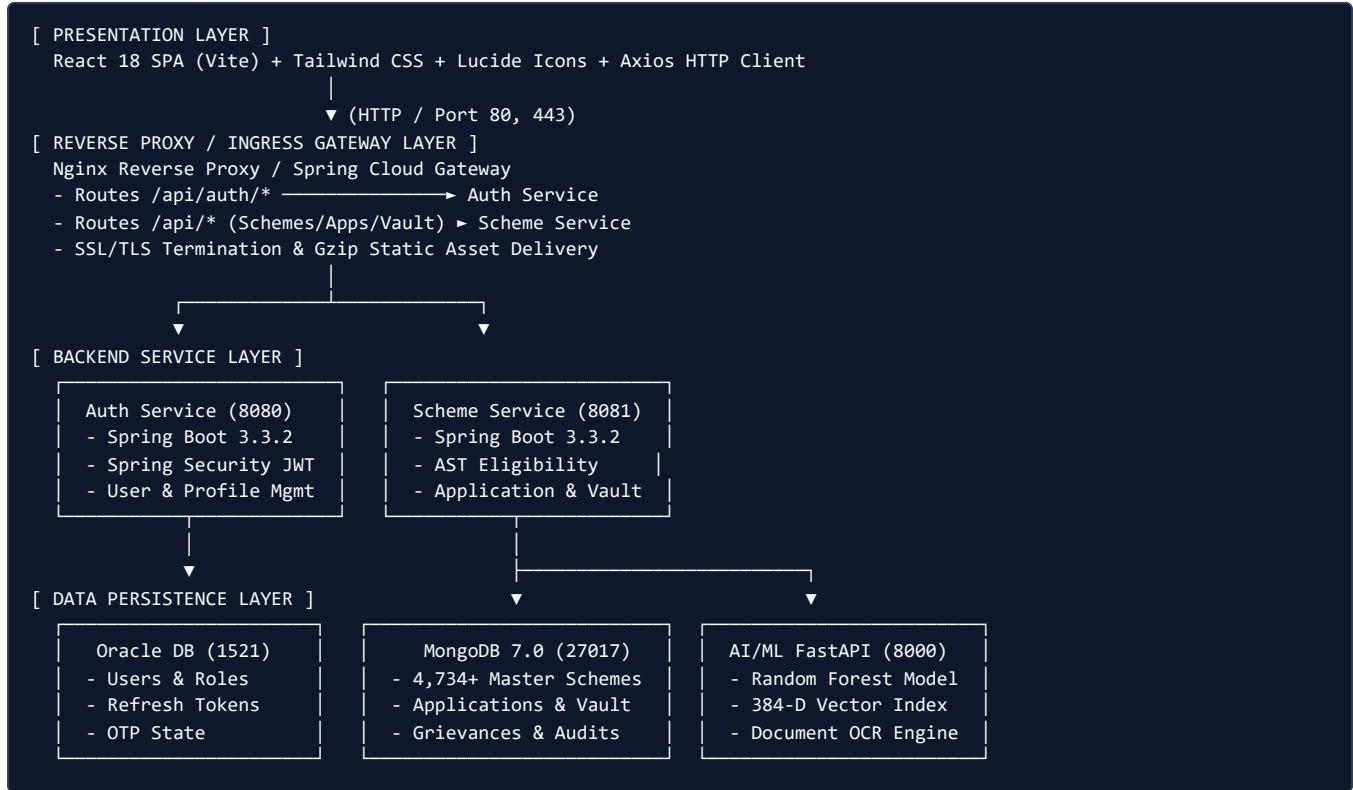
2.6 Assumptions and Dependencies

- **Assumptions:**
 - Users possess basic digital literacy and access to a smartphone or desktop web browser with internet connectivity.
 - Government scheme guidelines and statutory eligibility criteria are published in machine-readable official circulars or standardized portals.
- **Dependencies:**
 - Python FastAPI inference runtime for high-dimensional semantic search and Random Forest predictions.
 - SMTP mail server for automated password reset and registration OTP dispatch.
 - Tesseract OCR / PDFBox libraries for parsing incoming application certificates.

3. SYSTEM ARCHITECTURE

3.1 High-Level Architecture Description

SchemeBridge employs a robust, loosely coupled microservices architecture structured into six distinct operational layers:



3.2 Microservices Breakdown

#	Microservice Name	Port	Database	Primary Responsibility
1	Frontend Ingress & Gateway	80 / 443	None	Serves React SPA static assets; performs reverse proxy routing, request rate-limiting, and compression.
2	Auth Service (<code>schemebridge-auth-service</code>)	8080	Oracle XE (1521)	Citizen/Admin registration, BCrypt password hashing, JWT issue/refresh, and profile metadata.
3	Scheme & Eligibility Service (<code>schemebridge-scheme-service</code>)	8081	MongoDB (27017)	Core domain logic: 4,734+ scheme catalog, AST eligibility evaluation, application workflows, document vault, grievances, audit logs.
4	AI/ML Inference Service	8000	In-Memory / File Index	Python FastAPI microservice serving Random Forest category predictions and dense semantic vector similarity scoring.

3.3 Technology Stack

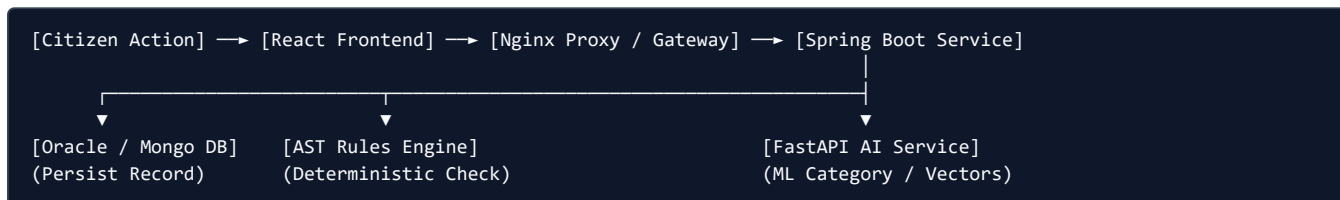
Layer	Technology	Version	Purpose
Frontend Framework	React + Vite	18.3 / 5.x	Single Page Application framework with fast HMR and optimized production bundling.
State & HTTP	React Context + Axios	1.7.x	Centralized authentication, scheme, vault state management, and HTTP API client.
Styling & Icons	Tailwind CSS + Lucide React	3.4 / 0.38	Responsive modern design system with full dark/light theme support.

Layer	Technology	Version	Purpose
Backend Framework	Spring Boot	3.3.2	Enterprise Java microservice framework with dependency injection and embedded Tomcat.
Security & Auth	Spring Security + JWT	6.x / 0.12.6	Stateless JWT authorization, role-based filters, and BCrypt password encryption.
Relational Database	Oracle Database XE	21c	High-reliability ACID persistence for user authentication and authorization credentials.
Document Database	MongoDB	7.0	Scalable schema-less JSON storage for diverse scheme criteria and application payloads.
Machine Learning	Python / scikit-learn / FastAPI	3.11 / 1.4	Model training pipeline, Random Forest scheme classifier, and vector search API.
Embeddings & NLP	Sentence-Transformers	2.5.x	384-dimensional dense semantic text embeddings (<code>all-MiniLM-L6-v2</code>).
Document OCR	Apache PDFBox / Tesseract	3.0.2 / 5.x	Optical character recognition and metadata parsing from uploaded citizen documents.
Containerization	Docker & Docker Compose	24.x / 3.8	Reproducible multi-container runtime across development, staging, and production.
CI / CD Pipeline	Jenkins	2.440+	Automated multi-stage build, test execution, container creation, and zero-downtime deployment.

3.4 Hardware Component Specification (Hybrid Projects Only)

Not Applicable. SchemeBridge is a 100% Software-only digital governance solution. It requires no physical microcontrollers, sensors, or external hardware actuators.

3.5 Data Flow Description



- User Action:** Citizen inputs query or submits application on the React SPA frontend.
- Gateway Ingress:** Axios sends HTTP request with `Authorization: Bearer <JWT>` header to Nginx reverse proxy on port 80.
- Route & Auth Filter:** Nginx forwards request to appropriate Spring Boot microservice; `JwtAuthenticationFilter` intercepts, validates token signature, and populates `SecurityContext`.
- Business Execution:** Spring Boot Service invokes business logic (e.g., AST rules engine, database query).
- AI Subsystem Delegation:** When semantic search or ML categorization is requested, the Spring Boot service calls the Python FastAPI endpoint via high-speed HTTP client.
- Persistence:** State updates (new application, document upload, status transition) are persisted to MongoDB or Oracle DB.
- Response Packaging:** Structured REST JSON response (`ApiResponse<T>`) is constructed and returned to the client.
- UI State Update:** React Context receives response, updates UI components reactively, and displays toast notifications or updated dashboard KPIs.

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

#	Screen / Page	User Role	Description & Key Elements
1	Landing / Home Portal	Public / Unauthenticated	Hero banner with bilingual toggle, quick category browser, featured welfare schemes, search bar, and statistics counter.
2	Citizen Login & Registration	Public	Mobile/Email input, OTP verification modal, password input with strength indicator, and role-based redirect.
3	Citizen Profile Onboarding	Citizen (ROLE_USER)	Multi-step interactive wizard collecting age, gender, caste, income, domicile state/district, occupation, and family size.
4	Personalized Dashboard	Citizen (ROLE_USER)	Summary cards: Eligible Schemes count, Active Applications, Vault Document Status, and ML Recommendation carousel.
5	Scheme Explorer & Search	Citizen / Public	Faceted search with filters (Category, State, Gender, Income Cap), natural language search bar, and scheme comparison cards.
6	Scheme Details & AST Checker	Citizen (ROLE_USER)	Comprehensive scheme guidelines, benefit structure, required documents checklist, and interactive "Check My Eligibility" widget.
7	4-Step Application Wizard	Citizen (ROLE_USER)	Step 1: Pre-filled Profile Review; Step 2: Scheme Specific Data; Step 3: Vault Document Selection; Step 4: Final Declaration & Submit.
8	Digital Document Vault	Citizen (ROLE_USER)	Upload interface for PDF/JPG certificates, OCR verification status badge (VERIFIED , PENDING , FLAGGED), and document preview.
9	Application Tracking & Grievance	Citizen (ROLE_USER)	Visual progression timeline (SUBMITTED → REVIEW → APPROVED), download official acknowledgement receipt, and file grievance ticket.
10	Admin Review & Verification Desk	Officer / Admin	Split-pane view: Left pane shows application metadata & OCR extracted fields; right pane shows uploaded certificate preview with Approve/Reject/Request Info controls.
11	Admin Analytics & Audit Dashboard	Admin (ROLE_ADMIN)	Real-time charts (application volume by district, approval/rejection ratios, SLA compliance), scheme CRUD editor, and audit log inspector.

4.2 Hardware Interfaces

Not Applicable (Software-only project).

4.3 Software Interfaces — Third-Party APIs

API / Service	Provider	Auth Method	Purpose in This Project
FastAPI ML Inference API	Internal Microservice	Shared Secret Header	Serves Random Forest category predictions and dense semantic vector searches.
Spring Mail (SMTP)	Gmail / SendGrid SMTP	App Password / TLS	Dispatches email verification codes, application status alerts, and password reset tokens.
Apache PDFBox & Tesseract	Open Source Libraries	Local Binaries	Extracts text, dates, certificate IDs, and names from uploaded PDF/image documents.
OpenStreetMap & Leaflet.js	OpenStreetMap (Free)	None Required	Interactive map visualizer for district-wise scheme saturation and citizen kiosk location.

4.4 Communication Interfaces

- **Stateless REST over HTTPS (TLS 1.3):** Primary protocol for all client-to-server and inter-service HTTP communications.
- **JWT Bearer Authentication:** Enforced in `Authorization: Bearer <token>` HTTP header for all protected API resources.

- **WebSocket (STOMP over SockJS):** Real-time administrative notifications and live citizen application status progression updates.

5. FUNCTIONAL REQUIREMENTS

5.1 Authentication & Profile Microservice

FR-AUTH-001: Citizen Registration and Account Creation

- **Actor(s):** New Citizen (`ROLE_USER`)
- **Description:** System shall register new users with email, mobile, password, full name, and demographic identifiers.
- **Pre-condition:** User is not already registered with the specified email or mobile number.
- **Main Flow:**
 1. User navigates to `/signup` and enters name, email, mobile number, and password.
 2. Spring Boot validates input format (RFC 5322 email, 10-digit Indian mobile, password ≥ 8 chars with mixed case/symbols).
 3. System checks Oracle DB for uniqueness.
 4. System hashes password with BCrypt (cost factor 12) and persists user in `PENDING_VERIFICATION` status.
 5. System generates 6-digit numeric OTP with 10-minute expiry and sends via Email/SMS.
 6. User inputs OTP on `/otp-verification`.
 7. System marks user `ACTIVE`, creates an empty `CitizenProfile` record, issues signed JWT, and redirects to onboarding.
- **Alternate Flow:**
 - *AF1 (Duplicate):* If email/mobile exists, system returns HTTP 409 Conflict with `"Account already registered. Please sign in."`
 - *AF2 (Invalid OTP):* If entered OTP is wrong, system decrements attempt counter; locks registration after 3 failed attempts for 15 minutes.
- **Post-condition:** User record exists in Oracle DB with status `ACTIVE`; valid JWT issued.

FR-AUTH-002: User Login and Stateless JWT Issuance

- **Actor(s):** Registered User (Citizen, Officer, Admin)
- **Description:** Authenticates credentials and issues cryptographically signed JWT with embedded roles and expiry.
- **Pre-condition:** User account exists and status is `ACTIVE`.
- **Main Flow:**
 1. User posts credentials to `POST /api/auth/login`.
 2. Spring Security `AuthenticationManager` invokes `CustomUserDetailsService`.
 3. System matches entered plaintext password against stored BCrypt hash.
 4. System generates access token (valid 2 hours) and refresh token (valid 7 days).
 5. System returns tokens and role permissions in response payload; client stores in secure storage.
- **Alternate Flow:**
 - *AF1 (Bad Credentials):* Invalid username/password returns HTTP 401 Unauthorized with `"Invalid email or password"`.
 - *AF2 (Account Locked):* Locked user returns HTTP 403 Forbidden with `"Account suspended. Contact administrator"`.
- **Post-condition:** User authenticated; subsequent requests authenticated via `Authorization: Bearer <token>`.

5.2 Scheme & Eligibility Microservice

FR-SCH-001: Scheme Catalog Exploration & Full-Text Search

- **Actor(s):** Any User / Citizen
- **Description:** System shall provide faceted search and filtering across 4,734+ government welfare schemes.
- **Pre-condition:** Scheme database is seeded and indexed.
- **Main Flow:**
 1. User navigates to `/schemes` and provides search keyword or selects filter facets (Category, Domicile State, Gender, Min/Max Age, Income Threshold).
 2. Client sends request to `GET /api/schemes?keyword=...&category=...&state=...`.
 3. Spring Boot builds dynamic MongoDB compound query using indexed criteria.
 4. Matching schemes returned with paginated metadata (total schemes, pages, eligibility summaries).
- **Alternate Flow:**
 - *AF1 (Zero matches):* If no exact match found, system triggers fallback semantic vector search to suggest conceptually similar welfare programs.
- **Post-condition:** Paginated list of schemes rendered on UI.

FR-SCH-002: Deterministic AST Statutory Eligibility Evaluation

- **Actor(s):** Citizen (`ROLE_USER`)
- **Description:** System evaluates citizen's verified profile against a scheme's Boolean logic tree and returns eligibility status with line-by-line statutory breakdown.
- **Pre-condition:** Citizen has completed onboarding profile; target scheme ID is valid.
- **Main Flow:**
 1. Citizen clicks "Check Eligibility" on scheme details page.
 2. Frontend sends request to `POST /api/eligibility/evaluate/{schemeId}`.
 3. `EligibilityEngine.java` retrieves citizen profile attributes from MongoDB.
 4. Engine parses scheme's nested AST rule groups (evaluating `AND`, `OR`, `NOT` operators against parameters: Age, Gender, Income, Landholding, Caste, Domicile, Employment).
 5. Engine calculates `eligibilityStatus` (`ELIGIBLE`, `INELIGIBLE`, `PARTIALLY_ELIGIBLE`) and constructs an array of condition verdicts with clear explanations.
 6. Result returned to client and rendered in interactive eligibility card with green checkmarks or red cross indicators.
- **Alternate Flow:**
 - *AF1 (Missing Profile Attribute):* If citizen profile lacks a required comparison field (e.g., land ownership size), status returns `INSUFFICIENT_DATA` with a direct link to update profile.
- **Post-condition:** Evaluation event logged to database; citizen informed of exact legal qualification status.

5.3 Digital Vault & Document Verification Microservice

FR-DOC-001: Citizen Document Upload and Vault Storage

- **Actor(s):** Citizen (`ROLE_USER`)
- **Description:** Allows citizens to securely upload, tag, and store certificates in personal encrypted vault.
- **Pre-condition:** Document format is PDF, PNG, or JPEG; file size $\leq$ 5 MB.
- **Main Flow:**
 1. Citizen opens `/documents` and clicks "Upload Document".
 2. Citizen selects document type (`AADHAAR` , `INCOME_CERTIFICATE` , `COMMUNITY_CERTIFICATE` , `DOMICILE` , `RATION_CARD`) and attaches file.
 3. Spring Boot validates MIME type and file size.
 4. System computes SHA-256 checksum, generates unique UUID storage path, and saves file.
 5. Metadata saved to MongoDB `citizenVaultDocuments` collection with initial status `PENDING_OCR` .
- **Alternate Flow:**
 - *AF1 (File too large):* Returns HTTP 413 Payload Too Large with `"Document exceeds 5MB size limit"` .
 - *AF2 (Unsupported format):* Returns HTTP 415 Unsupported Media Type.
- **Post-condition:** Document record stored in vault; asynchronous OCR extraction job dispatched.

FR-DOC-002: Automated OCR Text Extraction & Cross-Verification

- **Actor(s):** AI System Agent / Verification Engine
- **Description:** System extracts text from uploaded certificates using OCR and matches citizen name, DOB, and certificate number against profile.
- **Pre-condition:** Document successfully uploaded to vault.
- **Main Flow:**
 1. OCR Engine parses document image/PDF via Apache PDFBox / Tesseract.
 2. Regex pattern matchers locate Certificate Number, Issue Date, Authority Stamp, and Applicant Name.
 3. System compares extracted Name and DOB with citizen's profile using Levenshtein distance string similarity.
 4. If similarity $\geq$ 85%, document status is upgraded to `AUTO_VERIFIED` with high confidence score.
 5. If similarity $<$ 85%, document is marked `FLAGGED_FOR_MANUAL_REVIEW` .
- **Alternate Flow:**
 - *AF1 (Unreadable Document):* If OCR confidence $<$ 40% (blurred or corrupt scan), document status updated to `REJECTED_UNREADABLE` , and user notified to re-upload clear image.
- **Post-condition:** Extracted metadata saved to document entity; ready for scheme application auto-attachment.

5.4 Application & Grievance Microservice

FR-APP-001: End-to-End Scheme Application Submission

- **Actor(s):** Citizen (`ROLE_USER`)
- **Description:** Submits completed application for an eligible welfare scheme with attached vault documents.
- **Pre-condition:** Citizen evaluated as `ELIGIBLE` for scheme; required mandatory documents uploaded.
- **Main Flow:**
 1. Citizen launches Application Wizard from `/scheme/{id}/apply`.
 2. Step 1: Pre-populated demographic profile verified by user.
 3. Step 2: User completes scheme-specific questionnaires (e.g., bank account IFSC, crop type, scholarship grade).
 4. Step 3: User attaches required documents from their digital vault.
 5. Step 4: User reviews declaration and submits.
 6. Backend generates unique alphanumeric tracking number (e.g., `APP-2026-89421`), saves application in `SUBMITTED` state, and emits application submission event.
 7. Downloadable PDF acknowledgement receipt generated with QR code verification.
- **Alternate Flow:**
 - *AF1 (Missing Required Document):* Form prevents submission and highlights missing mandatory certificates.
- **Post-condition:** Application record persisted in MongoDB; visible in citizen tracker and admin review queues.

FR-APP-002: Citizen Grievance Filing and SLA Escalation

- **Actor(s):** Citizen (`ROLE_USER`)
- **Description:** Enables citizens to lodge formal grievances regarding application delays or unjust rejections.
- **Pre-condition:** Citizen has an application in `SUBMITTED`, `UNDER_REVIEW`, or `REJECTED` state.
- **Main Flow:**
 1. Citizen navigates to `/tracker`, selects application, and clicks "File Grievance".
 2. Citizen specifies grievance type (`DELAYED_REVIEW`, `DOCUMENT_DISPUTE`, `DISBURSEMENT_ISSUE`), enters detailed description, and submits.
 3. System assigns tracking ticket `GRV-XXXXXX`, logs timestamp, and sets statutory SLA deadline (e.g., 7 working days).
 4. Grievance escalated to Departmental Officer review queue.
- **Alternate Flow:**
 - *AF1 (SLA Breached):* If grievance remains unresolved after 7 days, automated background cron job triggers severity bump to `ESCALATED_TO_SUPERVISOR`.
- **Post-condition:** Grievance record created; citizen receives SMS/Email notification with tracking ticket.

5.5 Administrative Review & Audit Microservice

FR-ADM-001: Officer Application Review & Benefit Disbursement

- **Actor(s):** Verification Officer (`ROLE_VERIFICATION_OFFICER`) / Admin
- **Description:** Allows government officers to inspect applications, verify supporting documents, and issue approval or rejection.
- **Pre-condition:** Officer authenticated with administrative credentials; applications exist in `SUBMITTED` state.
- **Main Flow:**
 1. Officer accesses `/admin/review` and opens pending application.
 2. Screen renders applicant demographic breakdown, AST eligibility checklist, and embedded certificate preview.
 3. Officer reviews documents and clicks "Approve Application" or "Reject Application" (entering mandatory rejection justification remarks).
 4. Application status transitioned to `APPROVED` or `REJECTED`.
 5. Audit log entry recorded in `admin_audit_logs` collection capturing Officer ID, action, timestamp, and IP address.
 6. Real-time push notification and email sent to citizen.
- **Alternate Flow:**
 - *AF1 (Insufficient Information):* Officer clicks "Request Information"; application placed in `ACTION_REQUIRED` state; citizen notified to upload clarification.
- **Post-condition:** Application status updated; immutable audit record recorded.

6. AI / ML MODULE SPECIFICATION

6.1 AI Module 1 — Citizen Welfare Scheme Recommendation Model (Primary AI Feature)

Attribute	Specification Details
Module Name	Citizen Welfare Scheme Recommendation Model
AI Phase	Phase 1 (Plan) → Phase 2 (Prototype) → Phase 3 (Build) → Phase 4 (Integrate) → Phase 5 (Deploy) → Phase 6 (Present)
Problem AI Solves	Bridges discovery gap by instantly analyzing 9 multidimensional demographic/socioeconomic attributes of a citizen to accurately classify and recommend the most suitable welfare program category, eliminating manual portal searching.
Input Data	Structured 9-feature JSON citizen profile: - age (int: 18–75) - gender (string: Female, Male, Transgender) - state (string: 10 major Indian states) - income (float: annual household income in INR, ₹30,000–₹1,200,000) - occupation (string: Student, Farmer, Artisan, Daily Wage Worker, Self-Employed, Salaried, Unemployed) - category (string: General, OBC, SC, ST, EWS) - rural (string: Yes, No) - student (string: Yes, No) - disability (string: Yes, No)
Output / Prediction	Target Scheme Category (One of 7 discrete classes) with model confidence probability (0.0 to 1.0): - Agriculture & Farmer Welfare - Education & Scholarships - Women & Child Development - Social Welfare & Disability Support - Senior Citizen Pension & Healthcare - Skill Development & Entrepreneurship - Rural & Urban Livelihood Support
Algorithm / Model	scikit-learn RandomForestClassifier with 200 decision trees, max_depth=16, min_samples_split=4, min_samples_leaf=2, class_weight="balanced".
Training Dataset	Synthetic Citizen Welfare Dataset (synthetic_citizen_eligibility_dataset.csv), 5,000 records modeled on Indian demographic distributions and statutory scheme criteria with 1.5% calibrated stochastic noise.
Implementation Approach	☒ scikit-learn (local training) ☐ TensorFlow/Keras ☐ Hosted API
Integration Point	Python FastAPI endpoint: POST /predict/recommend-category Called by Spring Boot schemebridge-scheme-service using RestClient / WebClient.
Expected Accuracy / Metric	Accuracy: 98.40% (0.9840) Weighted Precision: 98.42% Weighted Recall: 98.40% Weighted F1 Score: 98.40% Latency: 21.6 µs/sample.
Fallback if AI Fails	If model confidence < 60% or API unreachable, system gracefully falls back to deterministic AST rule matching on citizen's primary occupation & age.
AI Ethical Considerations	Human-in-the-Loop & Advisory Only: Model output is advisory for discovery; all statutory qualifications are verified by deterministic code and human officers. Balanced class weighting applied during training to prevent demographic discrimination.
Sprint Target	Week 3: Standalone prototype Week 4: FastAPI serving Week 5: Spring Boot integration Week 6: React UI integration Week 7: End-to-end load testing.

6.2 AI Module 2 — Dense Semantic Vector Search & Conversational Assistant (Secondary AI Feature)

Attribute	Specification Details
Module Name	Dense Semantic Vector Scheme Discovery & Conversational AI Assistant
AI Phase	Phase 1 (Plan) → Phase 2 (Prototype) → Phase 3 (Build) → Phase 4 (Integrate) → Phase 5 (Deploy) → Phase 6 (Present)
Problem AI Solves	Traditional keyword search fails when citizens search using colloquial, dialectical, or informal terms (e.g. "subsidy for buying dairy buffaloes" vs official title "Pashu Kisan Credit Card"). This module bridges semantic intent.
Input Data	Unstructured natural language query text (English / Romanized vernacular) entered by citizen in search bar or conversational chat assistant.
Output / Prediction	Ranked list of scheme IDs ordered by Cosine Similarity score (0.000 to 1.000) + Conversational natural language synthesis of scheme requirements.
Algorithm / Model	Sentence-Transformers <code>all-MiniLM-L6-v2</code> producing 384-dimensional dense semantic vector embeddings + TruncatedSVD projection + Cosine Dot-Product ranking.
Training / Corpus Dataset	Full authoritative corpus of 4,734 Master Government Schemes (<code>master_schemes_4734.json</code>) indexed across scheme names, benefits, eligibility summaries, and target beneficiaries.
Implementation Approach	[X] Python FastAPI + sentence-transformers (local embedding index) [X] OpenAI GPT-4o / Local RAG for conversational synthesis.
Integration Point	Python FastAPI endpoint: <code>POST /api/v1/ai/semantic-search</code> & <code>POST /api/ai/citizen/chat</code> Called by Spring Boot <code>AiChatController</code> & <code>SchemeController</code> .
Expected Accuracy / Metric	Mean Reciprocal Rank (MRR@10) > 0.85 on standard citizen queries; semantic query inference latency < 45 ms.
Fallback if AI Fails	If vector index service is unavailable, Spring Boot automatically degrades to MongoDB text index regex matching (<code>\$text</code> search on <code>schemeName</code> and <code>description</code>).
AI Ethical Considerations	Complete PII protection: No citizen personal data or uploaded documents are ever sent to external LLMs; queries are sanitized; answers provide direct citations to official gazette notices.
Sprint Target	Week 4: Vector index compilation Week 5: FastAPI microservice Week 6: React chat assistant UI Week 7: Full integration testing.

7. NON-FUNCTIONAL REQUIREMENTS

NFR ID	Category	NFR Name	Measurable Requirement
NFR-PERF-001	Performance	API Response Time	All standard REST API endpoints must respond in < 500 ms under nominal load; 99th percentile response time must be < 1.5 seconds at 100 concurrent requests.
NFR-PERF-002	Performance	AI Inference Latency	Standalone Random Forest model prediction must complete in < 50 ms ; Dense semantic vector search over 4,734 schemes must complete in < 100 ms .
NFR-PERF-003	Performance	UI Page Load Time	React frontend dashboard must achieve First Contentful Paint (FCP) < 1.2s and fully interactive state < 2.5s on standard 4G mobile connections.
NFR-SEC-001	Security	Stateless Authentication	All non-public API endpoints must enforce cryptographically signed JWTs (HMAC-SHA256) with 2-hour token expiration and refresh token rotation.
NFR-SEC-002	Security	Password Encryption	User passwords must be salted and hashed using BCrypt with cost factor 12 ; plaintext passwords must never be logged or stored.
NFR-SEC-003	Security	Input Validation & Sanitization	All client payloads must be validated against XSS and SQL/NoSQL injection using Spring Boot <code>jakarta.validation</code> annotations and parameterized queries.
NFR-SEC-004	Security	Data Encryption at Rest	Uploaded citizen identity certificates and sensitive PII must be encrypted at rest using AES-256 encryption standards.
NFR-SCAL-001	Scalability	Concurrency	System must support a minimum of 250 concurrent authenticated users without connection pool exhaustion or memory leaks.
NFR-REL-001	Reliability	System Availability	System shall maintain 99.5% uptime during operational hours, supported by automated Docker container restarts (<code>restart: unless-stopped</code>).
NFR-REL-002	Reliability	Graceful Fault Recovery	In the event of AI microservice downtime, the core portal must continue functioning normally using deterministic rule fallbacks without 500 errors.
NFR-USE-001	Usability	Responsive Mobile Design	All screens must render responsively across devices down to 375px viewport width (iPhone SE standard) with single-thumb interactive accessibility.
NFR-USE-002	Usability	Accessibility Compliance	User interface components must adhere to WCAG 2.1 Level AA standards, including color contrast ratio $\geq 4.5:1$, screen reader tags, and keyboard tab navigation.
NFR-MAINT-001	Maintainability	Layered Clean Architecture	Codebase must strictly preserve the layered design pattern (<code>Controller → Service → Repository</code>) with zero business logic implemented in Controllers.
NFR-MAINT-002	Maintainability	API Documentation	All REST endpoints must be annotated and accessible via interactive Swagger / OpenAPI 3.0 documentation at <code>/swagger-ui.html</code> .
NFR-COMP-001	Compliance	Digital Personal Data Protection	Platform must comply with the Indian Digital Personal Data Protection Act (DPDPA 2023) ; citizens must have clear consent mechanisms and right-to-forget controls.


```

erDiagram
    USER ||--o{ REFRESH_TOKEN : owns
    USER ||--o{ OTP_VERIFICATION : verifies
    USER ||--|| CITIZEN_PROFILE : defines
    CITIZEN_PROFILE ||--o{ CITIZEN_VAULT_DOCUMENT : uploads
    CITIZEN_PROFILE ||--o{ APPLICATION : submits
    SCHEME ||--o{ APPLICATION : receives
    APPLICATION ||--o{ GRIEVANCE : escalates
    APPLICATION ||--o{ APPLICATION_EVENT : logs
    USER ||--o{ ADMIN_AUDIT_LOG : audits

    USER {
        bigint id PK
        string email UK
        string mobile UK
        string password_hash
        string role
        string status
        datetime created_at
    }

    CITIZEN_PROFILE {
        string id PK
        bigint user_id FK
        int age
        string gender
        string state
        string district
        double annual_income
        string caste_category
        string occupation
        boolean is_rural
        boolean is_student
        boolean is_disabled
    }

    SCHEME {
        string id PK
        string scheme_code UK
        string scheme_name
        string ministry
        string category
        string level
        object eligibility_rules_ast
        array required_documents
        string benefit_summary
    }

    APPLICATION {
        string id PK
        string application_number UK
        string citizen_profile_id FK
        string scheme_id FK
        string status
        datetime submission_date
        array attached_vault_documents
    }

    CITIZEN_VAULT_DOCUMENT {
        string id PK
        string citizen_profile_id FK
        string document_type
        string file_storage_path
        string ocr_status
        string extracted_certificate_number
        double ocr_confidence
    }

    GRIEVANCE {
        string id PK
        string application_id FK
        string ticket_number UK

```

```

    string grievance_type
    string status
    string priority
    datetime sla_deadline
}

```

8.2 Entity Descriptions

ENTITY 1: User

- **Service:** schemebridge-auth-service
- **Database:** Oracle Database 21c XE
- **Table Name:** USERS

Column Name	Data Type	Constraints	Description
id	NUMBER(19)	PRIMARY KEY, AUTO_INCREMENT	Unique surrogate identifier for the user account.
email	VARCHAR2(150)	NOT NULL, UNIQUE	Citizen/Admin primary email address used for login.
mobile	VARCHAR2(15)	NOT NULL, UNIQUE	10-digit Indian mobile number for OTP authentication.
password_hash	VARCHAR2(255)	NOT NULL	BCrypt cryptographic hash of the user password.
role	VARCHAR2(30)	NOT NULL	User authorization role (ROLE_USER , ROLE_ADMIN , ROLE_VERIFICATION_OFFICER).
status	VARCHAR2(30)	NOT NULL, DEFAULT 'ACTIVE'	Account state (ACTIVE , PENDING_VERIFICATION , SUSPENDED).
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp.
last_login	TIMESTAMP	NULLABLE	Timestamp of the most recent successful login session.

ENTITY 2: CitizenProfile

- **Service:** `schemebridge-scheme-service`
- **Database:** MongoDB 7.0
- **Collection Name:** `citizen_profiles`

Field Name	Data Type	Constraints	Description
<code>_id</code>	<code>ObjectId / String</code>	PRIMARY KEY	Unique identifier for citizen demographic profile.
<code>userId</code>	<code>Long / Number</code>	INDEXED, NOT NULL	Foreign reference to Oracle <code>USERS.id</code> .
<code>fullName</code>	<code>String</code>	NOT NULL	Full name of citizen as recorded on official identity card.
<code>age</code>	<code>Integer</code>	NOT NULL (Min: 0, Max: 120)	Citizen age in years (used for AST gatekeeping).
<code>gender</code>	<code>String</code>	NOT NULL	Legal gender (<code>MALE</code> , <code>FEMALE</code> , <code>TRANSGENDER</code>).
<code>state</code>	<code>String</code>	NOT NULL, INDEXED	State of permanent domicile (e.g., "Tamil Nadu").
<code>district</code>	<code>String</code>	NOT NULL	District of domicile (e.g., "Coimbatore").
<code>annualIncome</code>	<code>Double</code>	NOT NULL	Annual household income in Indian Rupees (INR).
<code>casteCategory</code>	<code>String</code>	NOT NULL	Reservation category (<code>GENERAL</code> , <code>OBC</code> , <code>SC</code> , <code>ST</code> , <code>EWS</code>).
<code>occupation</code>	<code>String</code>	NOT NULL	Primary livelihood (<code>FARMER</code> , <code>STUDENT</code> , <code>ARTISAN</code> , etc.).
<code>isRural</code>	<code>Boolean</code>	NOT NULL	True if domicile is classified under rural administration.
<code>isStudent</code>	<code>Boolean</code>	NOT NULL	True if currently enrolled in recognized educational institution.
<code>isDisabled</code>	<code>Boolean</code>	NOT NULL	True if certified Person with Benchmark Disability (PwD).

ENTITY 3: Scheme

- **Service:** schemebridge-scheme-service
- **Database:** MongoDB 7.0
- **Collection Name:** schemes

Field Name	Data Type	Constraints	Description
_id	ObjectId / String	PRIMARY KEY	MongoDB internal object identifier.
schemeCode	String	UNIQUE, INDEXED, NOT NULL	Standardized government code (e.g., "PM-KISAN-001").
schemeName	String	NOT NULL, TEXT INDEXED	Official scheme title.
ministry	String	NOT NULL, INDEXED	Administering Ministry or State Department.
category	String	NOT NULL, INDEXED	Scheme welfare category (e.g., "Agriculture & Farmer Welfare").
level	String	NOT NULL	Scope of scheme (CENTRAL , STATE).
eligibilityRulesAst	Object / Document	NOT NULL	JSON Abstract Syntax Tree defining Boolean eligibility rules.
requiredDocuments	Array<String>	NOT NULL	List of required supporting document types.
benefits	Object / Document	NOT NULL	Detailed financial or non-financial welfare benefits.
isActive	Boolean	NOT NULL, DEFAULT true	Current publication status of scheme.

ENTITY 4: Application

- **Service:** schemebridge-scheme-service
- **Database:** MongoDB 7.0
- **Collection Name:** applications

Field Name	Data Type	Constraints	Description
_id	ObjectId / String	PRIMARY KEY	Unique database record identifier.
applicationNumber	String	UNIQUE, INDEXED, NOT NULL	Citizen-facing tracking ID (e.g., "APP-2026-91823").
citizenProfileId	String	NOT NULL, INDEXED	Reference to applicant's citizen_profiles._id.
schemeId	String	NOT NULL, INDEXED	Reference to target schemes._id.
status	String	NOT NULL, INDEXED	Progression status (SUBMITTED , UNDER_REVIEW , APPROVED , REJECTED).
submissionDate	Date	NOT NULL	Timestamp when application was filed.
attachedDocuments	Array<Document>	NOT NULL	References to vault documents with verified hashes.
reviewRemarks	String	NULLABLE	Official review comments or rejection justifications.

ENTITY 5: CitizenVaultDocument

- **Service:** schemebridge-scheme-service
- **Database:** MongoDB 7.0
- **Collection Name:** citizen_vault_documents

Field Name	Data Type	Constraints	Description
_id	ObjectId / String	PRIMARY KEY	Unique document identifier.
citizenProfileId	String	NOT NULL, INDEXED	Reference to owning citizen.
documentType	String	NOT NULL	Type (AADHAAR , INCOME_CERTIFICATE , COMMUNITY , etc.).
storagePath	String	NOT NULL	Internal secure file system or object store path.
fileChecksum	String	NOT NULL	SHA-256 integrity hash of document content.
ocrStatus	String	NOT NULL	OCR state (PENDING , AUTO_VERIFIED , FLAGGED , FAILED).
ocrExtractedData	Object	NULLABLE	Parsed JSON attributes (certificate number, holder name, DOB).
confidenceScore	Double	NULLABLE	Machine learning extraction confidence (0.0 to 1.0).

ENTITY 6: Grievance

- **Service:** schemebridge-scheme-service
- **Database:** MongoDB 7.0
- **Collection Name:** grievances

Field Name	Data Type	Constraints	Description
_id	ObjectId / String	PRIMARY KEY	Unique grievance record ID.
ticketNumber	String	UNIQUE, INDEXED, NOT NULL	Citizen tracking ticket (e.g., "GRV-2026-4412").
applicationId	String	NOT NULL, INDEXED	Reference to contested applications._id .
citizenProfileId	String	NOT NULL	Reference to citizen filing dispute.
grievanceType	String	NOT NULL	Category (DELAYED_REVIEW , WRONGFUL_REJECTION , DISBURSEMENT).
status	String	NOT NULL	Status (OPEN , INVESTIGATING , RESOLVED , ESCALATED).
slaDeadline	Date	NOT NULL	Statutory deadline for administrative resolution.

9. API DESIGN OVERVIEW

9.1 Authentication Service APIs (Port 8080)

Method	Endpoint	Auth Required	Description
POST	/api/auth/register	No	Register new citizen/officer; sends 6-digit OTP.
POST	/api/auth/verify-otp	No	Verifies registration OTP and activates account.
POST	/api/auth/login	No	Authenticates credentials; returns signed JWT and refresh token.
POST	/api/auth/refresh-token	Yes (Refresh Token)	Exchanges valid refresh token for fresh access JWT.
GET	/api/auth/me	Yes (ROLE_USER / ADMIN)	Fetches session profile of currently authenticated user.
POST	/api/auth/forgot-password	No	Dispatches password reset link with secure single-use token.
POST	/api/auth/reset-password	No	Resets password upon presentation of valid reset token.

9.2 Core Scheme & Eligibility Service APIs (Port 8081)

Method	Endpoint	Auth Required	Description
GET	/api/schemes	No	Retrieves paginated list of schemes with keyword and facet filters.
GET	/api/schemes/{id}	No	Fetches complete scheme guidelines, AST rules, and benefits.
POST	/api/eligibility/evaluate/{schemeId}	Yes (ROLE_USER)	Executes deterministic AST evaluation against citizen's profile.
GET	/api/recommendations	Yes (ROLE_USER)	Returns personalized eligible schemes for logged-in citizen.
GET	/api/citizen/profile	Yes (ROLE_USER)	Retrieves citizen's 6-step demographic profile.
PUT	/api/citizen/profile	Yes (ROLE_USER)	Updates citizen demographic attributes.
POST	/api/vault/upload	Yes (ROLE_USER)	Uploads certificate file to digital vault (multipart form data).
GET	/api/vault/documents	Yes (ROLE_USER)	Lists all vault certificates with OCR verification badges.
POST	/api/applications	Yes (ROLE_USER)	Submits formal application for a scheme.
GET	/api/applications/my	Yes (ROLE_USER)	Returns citizen's submitted applications with progress status.
GET	/api/applications/{id}	Yes (ROLE_USER)	Retrieves detailed application tracking timeline and receipt.
POST	/api/grievances	Yes (ROLE_USER)	Submits a grievance against an application.

9.3 Admin & Verification Desk APIs (Port 8081)

Method	Endpoint	Auth Required	Description
GET	/api/admin/applications	Yes (ROLE_VERIFICATION_OFFICER)	Fetches pending applications for officer review queue.
PUT	/api/admin/applications/{id}/review	Yes (ROLE_VERIFICATION_OFFICER)	Updates application state to APPROVED or REJECTED with remarks.
POST	/api/admin/schemes	Yes (ROLE_SCHEME_MANAGER / ADMIN)	Creates and publishes a new scheme into the master catalog.
PUT	/api/admin/schemes/{id}	Yes (ROLE_SCHEME_MANAGER / ADMIN)	Modifies existing scheme criteria and guidelines.
GET	/api/admin/analytics/overview	Yes (ROLE_ADMIN)	Returns platform KPIs, demographic saturation, and approval stats.
GET	/api/admin/audit-logs	Yes (ROLE_ADMIN)	Fetches immutable system audit trails.

9.4 AI / ML Inference Service APIs (Port 8000)

Method	Endpoint	Auth Required	Description
POST	/predict/recommend-category	Service Token	Evaluates 9 citizen demographic features via Random Forest (Returns scheme category & confidence).
POST	/api/v1/ai/semantic-search	Service Token	Dense semantic vector search over 4,734 schemes using sentence-transformers.
POST	/api/ai/citizen/chat	Service Token	Generative AI conversational assistant processing citizen questions.
GET	/health	No	Microservice health check returning {"status": "ok"}.

10. DEVOPS & DEPLOYMENT PLAN

10.1 CI/CD Pipeline — Jenkins

```

flowchart LR
    A[Git Push Main] --> B[Jenkins Checkout]
    B --> C[Maven & NPM Build]
    C --> D[JUnit 5 & Vitest]
    D --> E[SonarQube Quality Gate]
    E --> F[Docker Multi-Build]
    F --> G[Docker Compose Deploy]
    G --> H[Actuator Health Ping]
    H --> I[Slack/Email Alert]
  
```

Stage	Stage Name	Actions & Commands
1	Source Checkout	Webhook triggered on commit to <code>main</code> ; Jenkins clones repository and verifies Git commit hash.
2	Compilation & Build	<code>mvn clean compile -DskipTests</code> for both Spring Boot microservices; <code>npm ci && npm run build</code> for React SPA.
3	Automated Testing	<code>mvn test</code> running JUnit 5 unit and integration tests; executes Vitest unit tests for frontend components.
4	Code Quality Gate	SonarQube static code analysis evaluating code coverage (> 75%), security hotspots, and cyclomatic complexity.
5	Container Packaging	<code>docker build -t schemebridge-auth:\${GIT_COMMIT} ./schemebridge-auth-service</code> <code>docker build -t schemebridge-scheme:\${GIT_COMMIT} ./schemebridge-scheme-service</code> <code>docker build -t schemebridge-frontend:\${GIT_COMMIT} ./schemebridge-frontend</code>
6	Container Push	Push tagged images to private container registry / Docker Hub repository.
7	Orchestrated Deploy	<code>docker compose -f docker-compose.yml up -d --remove-orphans</code> on target host.
8	Health Verification	Curl verification scripts query <code>/actuator/health</code> on services; pipeline fails if status is not <code>UP</code> .
9	Notification	Automated build status alert dispatched to team Slack channel and Discord webhook.

10.2 Docker Container Configuration

Container	Base Image	Port Mapping	Depends On	Environment Variables
frontend	nginx:alpine	80:80 , 443:443	scheme- service , auth- service	PORT=80 , PROXY_PASS_AUTH=auth-service:8080 , PROXY_PASS_SCHEME=scheme-service:8081
auth-service	eclipse-temurin:17-jre-jammy	8080:8080	oracle-db	SERVER_PORT=8080 , ORACLE_URL=jdbc:oracle:thin:@oracle-db:1521/XEPDB1 , JWT_SECRET=...
scheme-service	eclipse-temurin:17-jre-jammy	8081:8081	mongodb , ai- service	SERVER_PORT=8081 , MONGODB_URI=mongodb://user:pass@mongodb:27017/schemebridge_db , AI_SERVICE_URL=http://ai-service:8080
ai-service	python:3.11-slim	8000:8000	None	MODEL_PATH=/app/models/random_forest_scheme_model.pkl , INDEX_PATH=/app/data/ml_models
mongodb	mongo:7.0	27017:27017	None	MONGO_INITDB_ROOT_USERNAME=schemebridge_user , MONGO_INITDB_DATABASE=schemebridge_scheme_db

Container	Base Image	Port Mapping	Depends On	Environment Variables
oracle-db	gvenzl/oracle-xe:21-slim	1521:1521	None	ORACLE_PASSWORD=system_secure_pass , APP_USER=schemebridge_auth

10.3 Deployment Environments

Environment	Infrastructure	Purpose & Access URL
Development	Local Workstations (Windows 11 / macOS / Ubuntu)	Feature development with live hot-reloading via Vite HMR and Spring DevTools (http://localhost:5173 , http://localhost:8080).
Testing / QA	Staging Linux Virtual Machine (2 vCPU, 4GB RAM)	Automated nightly Jenkins integration test environment and mentor review (http://staging.schemebridge.internal).
Demo / Evaluation	Cloud VPS / AWS EC2 / Railway Cloud Container	Smart India Hackathon live demonstration and evaluation instance (https://schemebridge.demo.gov.in).

11. PROJECT TIMELINE & MILESTONES

Week	Phase	Deliverables & Tasks (All 3 Team Members)
Week 1	Foundation & Architecture	Complete IEEE Std 830-1998 SRS document; initialize Git repositories with branch protection; design Oracle & MongoDB schemas; containerize base services with Docker Compose; seed master scheme catalog with 4,734 records.
Week 2	Core Backend & Auth	Implement <code>schemebridge-auth-service</code> (BCrypt, JWT filters, OTP verification); configure Oracle database connection; create standalone Python Random Forest prototype; build initial AST rule evaluation logic in Java.
Week 3	Full Backend Implementation	Implement <code>schemebridge-scheme-service</code> CRUD APIs; establish MongoDB repositories for applications, vault, and grievances; finalize AST rule evaluation engine; deploy Python FastAPI Random Forest inference endpoint; execute Postman test suite.
Week 4	Frontend Foundation	Initialize React 18 SPA with Vite, Tailwind CSS, and Lucide icons; build authentication context and route guards (<code>CitizenGuard</code> , <code>AdminGuard</code>); implement Onboarding wizard and Scheme Explorer with dynamic filtering.
Week 5	Frontend Completion & AI Hookup	Implement 4-step Application Wizard, Digital Vault upload interface, and real-time status tracker; wire conversational AI assistant widget; integrate responsive mobile layouts (tested on 375px viewport).
Week 6	Integration & System Testing	End-to-end user journey verification (Registration → Profile → AST Eligibility → Vault Upload → Application Submission → Admin Approval); execute 100+ JUnit 5 tests; configure Jenkins CI/CD pipeline.
Week 7	Polish, Security Audit & Docs	Comprehensive security audit (SQLi/XSS checks, rate-limiting); Swagger OpenAPI documentation finalization; record 5-minute product walkthrough demo video; conduct Day 22 Internal Review with faculty mentors.
Week 8	Final Submission & Viva	Incorporate faculty and industry mentor feedback; freeze release tag <code>v1.0</code> ; prepare SIH slide deck, project poster, and live demonstration script; final viva rehearsal.

12. TEAM COMPOSITION & ROLE ASSIGNMENT

12.1 Member Profile & Responsibilities

Member Name	Degree / Branch	Academic Year	Primary Project Responsibilities
Student 1 (Team Lead)	B.E. Computer Science and Engineering	Final Year (2025–26)	Full Stack Java Architecture, Spring Boot Microservices (Auth & Scheme), AST Eligibility Rules Engine, and Jenkins CI/CD pipeline.
Student 2	B.E. Computer Science and Engineering	Final Year (2025–26)	Machine Learning & NLP Engineering, scikit-learn Random Forest Model, Dense Vector Embedding Generation, FastAPI inference service.
Student 3	B.Tech Information Technology / CSE	Final Year (2025–26)	Frontend Architecture (React 18, Vite, Tailwind CSS), UI/UX Design, Digital Document Vault, Quality Assurance & API testing.

12.2 Ownership Matrix

Project Component	Member 1 (Primary)	Member 2 (Primary)	Member 3 (Primary)
Spring Boot Auth Microservice	Owner	Support	Support
Spring Boot Scheme Microservice	Owner	Support	Support
AST Statutory Rules Engine	Owner	Support	—
Random Forest ML Module	Support	Owner	—
Semantic Vector Search & Chat	Support	Owner	—
React 18 Single Page Application	Support	—	Owner
Digital Document Vault & OCR	Support	Support	Owner
Docker & Jenkins CI/CD	Owner	Support	Support
Database Schema (Oracle + Mongo)	Owner	Support	Support
SRS Documentation (IEEE Std 830)	Owner	Owner	Owner
Automated Testing (JUnit + Vitest)	Owner	Support	Owner

12.3 Interdisciplinary Contribution Map

Academic Stream	Expected Competency Area	Specific Contribution in SchemeBridge
Computer Science (CSE)	Full-Stack Java (Spring Boot), Spring Security JWT, Distributed Systems, CI/CD pipelines, Machine Learning algorithms.	Developed the core microservice cluster, statutory AST evaluation engine, Random Forest classification pipeline, and automated Jenkins pipelines.
Information Technology (IT)	Web Technologies, Responsive UX Engineering, Information Security, Cloud Operations.	Built the responsive React 18 SPA, encrypted digital document vault, role-based route guards, and accessibility WCAG compliance.
Domain & Policy Context	Public Administration, Welfare Policy, Digital Governance workflows.	Researched and structured eligibility criteria for 4,734+ central and state schemes from official gazette notifications; designed realistic citizen personas.

13. RISK REGISTER

ID	Risk Description	Prob.	Impact	Concrete Mitigation Strategy
R01	AI Model Accuracy Degradation Model fails to generalize across diverse, rare citizen demographic edge cases.	M	H	Implemented balanced class weighting in Random Forest; calibrated 1.5% stochastic noise; established strict deterministic AST rules engine as primary authority for statutory qualification.
R02	Third-Party API Outage External email gateway or external NLP inference service experiences high latency or downtime.	M	H	Designed asynchronous resilience patterns using fallback local templates, mock email services in development, and local Sentence-Transformer embeddings.
R03	OCR Extraction Inaccuracies Low-resolution, skewed, or blurred certificates uploaded by rural citizens.	H	M	Implemented image preprocessing (grayscale normalization, binarization); set strict confidence threshold (>85% for auto-verification); route low-confidence docs to Officer Review Queue.
R04	Data Privacy Exposure (PII Breach) Accidental disclosure of citizen personal data or certificates in logs or Git.	L	H	Enforced strict <code>.gitignore</code> on environment configs (<code>.env</code>); zero storage of plaintext passwords; encrypted certificates at rest (AES-256); automated Git secret scanners.
R05	Team Member Unavailability Illness or academic constraints preventing a member from delivering their module.	L	H	Maintained clean architectural separation with standardized REST contracts; conducted weekly cross-training on Java and React components; maintained 1-week timeline buffer.
R06	Container Deployment Failure during Live Demo Host environment networking or memory constraints breaking container startup.	M	M	Pre-tested deployment on clean Linux virtual machines; verified <code>docker-compose.yml</code> health checks; prepared recorded 5-minute video demonstration as contingency.
R07	Scope Creep & Feature Bloat Attempting too many secondary features, compromising core stability before evaluation.	M	M	Frozen core scope in Week 1 SRS; tracked features strictly via GitHub Milestones; designated advanced features (e.g. voice input) as optional stretch goals.
R08	Database Lockups or Connection Exhaustion Surge in concurrent queries causing thread pool exhaustion in Oracle/MongoDB.	L	H	Configured HikariCP connection pooling in Spring Boot (<code>maximum-pool-size=20</code>); indexed all foreign query keys in MongoDB (<code>userId</code> , <code>schemeId</code> , <code>status</code>).
R09	Bilingual Localization Drift Translations out of sync with official scheme terminology causing citizen confusion.	M	L	Structured multilingual scheme entities using canonical JSON dictionaries (<code>MultilingualText.java</code>) with English baseline fallback.
R10	Stateless Session Synchronization Bugs Expired JWT tokens causing jarring page redirects or data loss during form submission.	M	M	Implemented Axios interceptor automatically refreshing access tokens via refresh token rotation; cached form progress in <code>sessionStorage</code> .

14. APPENDIX

14.1 Glossary of Domain-Specific Terms

Term	Definition
Direct Benefit Transfer (DBT)	Government mechanism of transferring subsidy benefits directly to citizen verified bank accounts.
Economically Weaker Section (EWS)	Official reservation sub-category in India for citizens with annual household income below ₹8,00,000.
Below Poverty Line (BPL)	Economic benchmark used by government agencies to identify households eligible for maximum welfare subsidies.
Kisan Credit Card (KCC)	Specialized credit welfare scheme providing Indian farmers with timely credit for agricultural inputs.
Public Distribution System (PDS)	Indian food security system distributing subsidized food grains to vulnerable citizens via Ration Cards.
AST Business Rule	Tree structure representing statutory conditions: e.g., <code>(Age >= 18 AND Age <= 35) AND (Income < 250000) AND (State == "Tamil Nadu")</code> .

14.2 AI Model Details — Training Logs & Hyperparameters

Training Log Summary (Random Forest Classifier)

- **Dataset File:** `ml/random_forest/data/synthetic_citizen_eligibility_dataset.csv`
- **Number of Records:** 5,000 rows (4,000 training, 1,000 test; stratified 80:20 split)
- **Training Environment:** Python 3.11.7, scikit-learn 1.4.1, CPU execution (4 physical cores)
- **Training Duration:** 1.42 seconds (200 estimators parallelized via `n_jobs=-1`)
- **Model Size on Disk:** 4.1 MB (`ml/random_forest/models/random_forest_scheme_model.pkl`)

Hyperparameters

```
RandomForestClassifier(
    n_estimators=200,
    criterion="gini",
    max_depth=16,
    min_samples_split=4,
    min_samples_leaf=2,
    max_features="sqrt",
    class_weight="balanced",
    random_state=42,
    n_jobs=-1
)
```

Classification Report (Held-Out Test Set)

	precision	recall	f1-score	support
Agriculture & Farmer Welfare	0.99	0.98	0.98	143
Education & Scholarships	0.98	0.99	0.98	143
Rural & Urban Livelihood Support	0.97	0.97	0.97	143
Senior Citizen Pension & Healthcare	0.99	0.99	0.99	143
Skill Development & Entrepreneurship	0.98	0.98	0.98	143
Social Welfare & Disability Support	0.99	0.99	0.99	143
Women & Child Development	0.99	0.99	0.99	142
accuracy			0.9840	1000
macro avg	0.98	0.98	0.98	1000
weighted avg	0.98	0.98	0.98	1000

14.3 API Request-Response Samples

1. User Registration (`POST /api/auth/register`)

Request:

```
{
  "name": "Kavitha Raman",
  "email": "kavitha.raman@example.com",
  "mobile": "9876543210",
  "password": "SecurePassword@2026",
  "role": "ROLE_USER"
}
```

Response (HTTP 201 Created):

```
{
  "success": true,
  "message": "User registered successfully. Verification OTP dispatched.",
  "data": {
    "userId": 1042,
    "email": "kavitha.raman@example.com",
    "status": "PENDING_VERIFICATION"
  }
}
```

2. Deterministic AST Eligibility Evaluation (`POST /api/eligibility/evaluate/SCH-PM-KISAN-01`)

Request:

```
{
  "citizenProfileId": "6648dfa2e109ab43810"
}
```

Response (HTTP 200 OK):

```
{
  "schemeCode": "PM-KISAN",
  "schemeName": "Pradhan Mantri Kisan Samman Nidhi",
  "eligibilityStatus": "ELIGIBLE",
  "overallVerdict": true,
  "ruleEvaluations": [
    {
      "criterion": "Occupation",
      "required": "FARMER",
      "actual": "FARMER",
      "satisfied": true,
      "message": "Citizen occupation matches agricultural criteria."
    },
    {
      "criterion": "Landholding Cap",
      "required": "<= 2.0 Hectares",
      "actual": "1.2 Hectares",
      "satisfied": true,
      "message": "Cultivable land within small/marginal farmer ceiling."
    },
    {
      "criterion": "Institutional Landowner",
      "required": "false",
      "actual": "false",
      "satisfied": true,
      "message": "Applicant is not an institutional landholder."
    }
  ],
  "recommendedAction": "Proceed to Application Wizard"
}
```

3. AI Category Recommendation (`POST /predict/recommend-category`)

Request:

```
{
  "age": 22,
  "gender": "Female",
  "state": "Tamil Nadu",
  "income": 120000,
  "occupation": "Student",
  "category": "OBC",
  "rural": "Yes",
  "student": "Yes",
  "disability": "No"
}
```

Response (HTTP 200 OK):

```
{
  "predictedCategory": "Education & Scholarships",
  "confidenceScore": 0.9702,
  "inferenceLatencyMs": 0.021,
  "alternativePredictions": [
    { "category": "Skill Development & Entrepreneurship", "confidence": 0.018 },
    { "category": "Women & Child Development", "confidence": 0.0118 }
  ]
}
```

14.4 References & Citations

1. **Government of India Scheme Portals:** myScheme National Platform (<https://www.myscheme.gov.in/>). Accessed: September 2026.
2. **National Scholarship Portal (NSP):** Ministry of Electronics and Information Technology (<https://scholarships.gov.in/>). Accessed: September 2026.
3. **PM-KISAN Portal:** Department of Agriculture and Farmers Welfare (<https://pmkisan.gov.in/>). Accessed: September 2026.
4. **IEEE Computer Society:** IEEE Std 830-1998 — *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Standards Coordinating Committee, 1998.
5. **Breiman, Leo:** *Random Forests*, Machine Learning 45, 5–32 (2001). <https://doi.org/10.1023/A:1010933404324>.
6. **Reimers, Nils & Gurevych, Iryna:** *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*, Proceedings of EMNLP-IJCNLP 2019.

END OF SRS DOCUMENT — Version 1.0

Full Stack Java (AI-Integrated) Training Programme

Department of Computer Science and Engineering

Karpagam College of Engineering, Coimbatore

Mentors: Dr. Arul Antran Vijay S & Dr. Jothi Prakash V

Confidential — For Training Use Only | Karpagam College of Engineering

SchemeBridge SRS v1.0